Telemetry and Final Configuration

How the completed build handled control-only ELRS, MAVLink-over-ELRS, optional DroneBridge Wi-Fi telemetry, and final configuration capture for repo handoff.

Telemetry options used in the project

- Option 1 — standard ELRS / CRSF: simplest control-only mode. The radio link carries pilot commands and basic signal-status feedback.
- Option 2 — MAVLink over ELRS: uses the ELRS path for both control and telemetry. Useful in the air, but slower for ground configuration than a direct USB cable or DroneBridge.
- Option 3 — DroneBridge on ESP32: preferred for on-ground Wi-Fi telemetry and fast access to a ground station without relying on the slower ELRS telemetry path.

Parameter profiles for each option

Mode	ArduPilot serial settings	ELRS TX setting	ELRS RX setting	Notes
CRSF control only	SERIAL6_BAUD = 460 SERIAL6_PROTOCOL = 23 RC_PROTOCOLS = 1	Link Mode = NORMAL	Serial Protocol = CRSF	Uses the serial line for normal ELRS control-link operation
MAVLink over ELRS	SERIAL6_BAUD = 460 SERIAL6_PROTOCOL = 2 RC_PROTOCOLS = 1	Link Mode = MAVLINK Backpack enabled on TX	Serial Protocol = MAVLINK	Used when the ELRS link carries both control and MAVLink telemetry
DroneBridge ESP32	Selected FC UART set to MAVLink2 at 57600 baud	Not required	Not required	Connect ground station to the ESP32 Wi-Fi telemetry bridge

DroneBridge setup sequence

- Flash the ESP32-C3 with the DroneBridge firmware using the web installer, then power-cycle the board.
- Connect to the access point named DroneBridge for ESP32 (password dronebridge) and open dronebridge.local.
- Set the UART serial protocol to MAVLink and the UART baud to 57600, then save and reboot.
- Wire the selected FC telemetry UART to the ESP32 and confirm that Mission Planner or QGroundControl can connect over UDP.

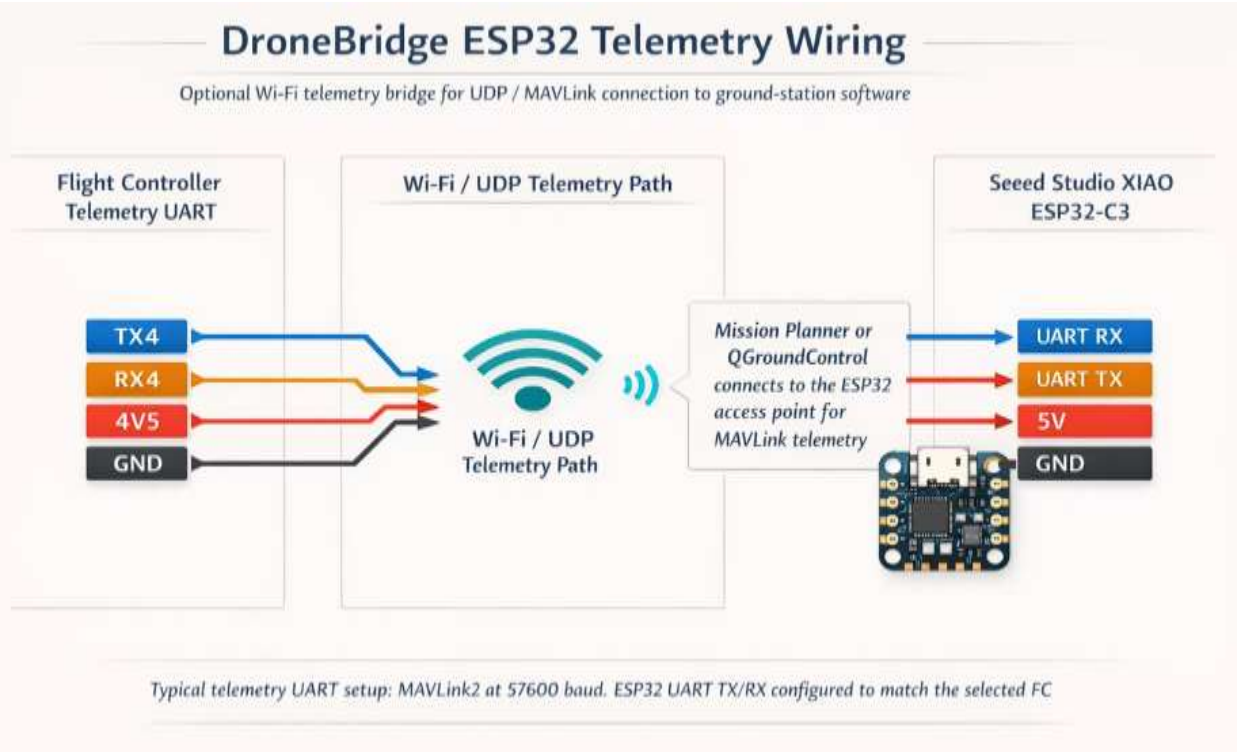

Figure 1. Telemetry hardware path used for the ESP32 Wi-Fi bridge.

Configuration baseline used in this repo package

Software / tool	Role in the build	Version used
ArduCopter	Flight-controller firmware	4.6.3
BLHeli_S	Motor-control firmware on ESCs	BLHeli_S
ExpressLRS	Radio-control link	4.0.0
DroneBridge for ESP32	Wi-Fi telemetry bridge	2.1.0
EdgeTX	Radio UI on transmitter	2.11.5
Mission Planner	Ground control and parameter setup	1.3.83
EdgeTX Companion	Radio model configuration	2.11.4
ELRS Backpack	Radio Wi-Fi telemetry feature	1.5.5
ExpressLRS Configurator	Radio / RX flashing utility	1.7.11